

A PROJECT TITLE

ON

Player Performance Prediction Using Regression and Gradient
Boosting

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE

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1. Abstract:

Cricket has evolved into a data-rich sport, with detailed ball-by-ball records now available for nearly every professional match. This project explores how such granular event data can be leveraged to predict individual player performance using machine learning. The Indian Premier League (IPL), spanning 2008–2023, was chosen as the dataset domain due to its consistent formatting, high match volume, and rich contextual detail — including venue, innings phase, and dismissal information at the delivery level.

The raw data, structured as one row per ball, was transformed through a feature engineering pipeline into a player-match-level dataset. Batting metrics (runs scored, strike rate, boundary counts) and bowling metrics (wickets, economy rate) were aggregated per player per match. To make the dataset suitable for predictive modeling, career-to-date rolling statistics were computed for each player, calculated exclusively from matches preceding the one being predicted. This time-aware construction was essential to prevent data leakage, ensuring that no feature relied on information unavailable at actual prediction time — a critical consideration for any sequential, real-world forecasting problem.

A Gradient Boosting Regressor was selected as the predictive model, given its strength in handling structured, tabular data with nonlinear relationships and mixed-scale features. The dataset was split chronologically rather than randomly, training on earlier matches and evaluating on more recent ones, to simulate a realistic forecasting scenario. The baseline model, using career-average runs, strike rate, and match experience as inputs, achieved a Mean Absolute Error of approximately 15.5 runs and a Root Mean Squared Error of approximately 20.7 runs, with career average runs emerging as the dominant predictive feature.

This baseline establishes a functional foundation for player performance prediction and highlights clear directions for improvement, including the incorporation of recent-form indicators, opponent-specific statistics, and venue effects. The project demonstrates an end-to-end machine learning workflow: data acquisition, feature engineering with leakage prevention, model training, and performance evaluation — applicable broadly to sports analytics and other time-dependent prediction problems.

2. INTRODUCTION :

Cricket generates rich ball-by-ball data, offering deep insight into player performance beyond simple averages. This project builds a machine learning pipeline to predict individual batting performance in IPL matches (2008–2023) using historical delivery-level data. Player features such as career-to-date average runs and strike rate were engineered while carefully avoiding data leakage — using only information available prior to each match. A Gradient Boosting Regression model was trained on this data, with performance evaluated using a chronological train-test split, establishing a baseline for player performance prediction in cricket analytics.

3. SOFTWARE REQUIREMENTS:

Programming Language

- Python 3.14

Development Environment

- Visual Studio Code (VS Code)

Version Control

- Git
- GitHub (remote repository hosting)

Python Libraries

- pandas — data loading, cleaning, and aggregation
- scikit-learn — machine learning model (Gradient Boosting Regressor), train-test splitting, evaluation metrics
- NumPy (used internally by pandas/scikit-learn for numerical computation)

Data Source & Access

- Kaggle (dataset repository)
- Kaggle API / CLI — for programmatic dataset download

Operating System

- Windows

Package Management

- pip (Python package installer)